SALES FORECASTING USING A STACKED ENSEMBLE MODEL

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success through innovation

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1.0 Abstract

Zimbabwean retail businesses face a challenging economic landscape characterized by volatility and a heavy reliance on imports. Intuition-based planning methods prevalent in the sector often lead to significant losses due to overstocking or understocking. I propose a sales forecasting system that uses a stacked ensemble machine learning model to forecast future sales to address this challenge. The system leverages historical sales data to identify trends, seasonality, and patterns. Stacked ensemble models combine the strengths of multiple models, leading to improved generalizability and robustness. By providing accurate sales predictions, this system empowers businesses to anticipate demand, identify growth opportunities, and mitigate risks associated with economic volatility. This research contributes to the field of sales forecasting by demonstrating the effectiveness of stacked ensemble models in a real-world, economically volatile environment, particularly for Zimbabwean retail businesses.

2.0. Introduction

The Zimbabwean retail industry despite its huge profitability in the country relies on instinct based decision making to plan market decisions to use. Despite its gross usage, it has its perks considering the currently volatile economic environment in the country which brings about unpredictability in the flow of the economic scale. As a result, many retail businesses suffer huge losses and even go bankrupt. To address this issue, I propose a sales forecasting system using a stacked ensemble model to predict future sales based on historical sales data and the prevailing market and economic conditions. The ensemble models are trained on historical sales and data and the predictions from each of the algorithms are combined through stacking to produce a more robust forecasting model. The model is then integrated to a web application for interactive usability by the user. The web application then accepts user input for the current sales and features that affect sales then the system processes the input and forecasts future values for the specified duration.

3.0. Problem definition

The Zimbabwean retail industry, a significant contributor to the national economy, suffers from substantial financial losses due to inefficiencies in inventory management and marketing strategies. These inefficiencies largely stem from a reliance on intuition-based decision making in a highly competitive and economically volatile environment. Traditional planning methods fail to account for critical factors like import dependency, fluctuating exchange rates, inflation, and seasonal trends. This results in overstocking (leading to losses of $471.9 billion globally) or understocking (costing $34.1 billion globally , Mohammad Ali (2023)), hindering profitability. Local Zimbabwean retail businesses lack the necessary data-driven strategies to optimize inventory management, resource allocation, and marketing campaigns in this challenging landscape.

This highlights the critical need for a robust and accurate sales forecasting system specifically tailored to the Zimbabwean retail sector. Such a system would empower businesses to make informed decisions, mitigate risks associated with economic volatility, and achieve sustainable growth.

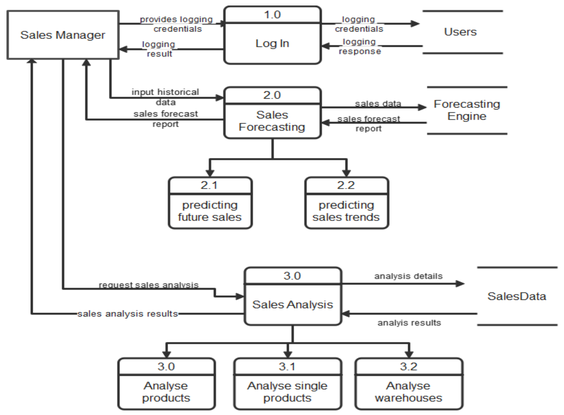
4.0. Overview of the developed application

To develop the sales forecasting system, a sales dataset with historical sales was trained using a stacked ensemble model of LightGBM, SARIMA (base-learners) and Random Forests,(meta-learner). The predictions of the two base-learners together with the original features are made inputs to the Random Forest meta learner using the stacking method to produce the final sales forecast. A web application will then be integrated with the pretrained model in order to create an interactive way for users to access the functionality of the forecasting model. Users log in to the system to access all the functionalities of the system. Users ill then have access to sales analysis functionalities and sales forecasting functionality.

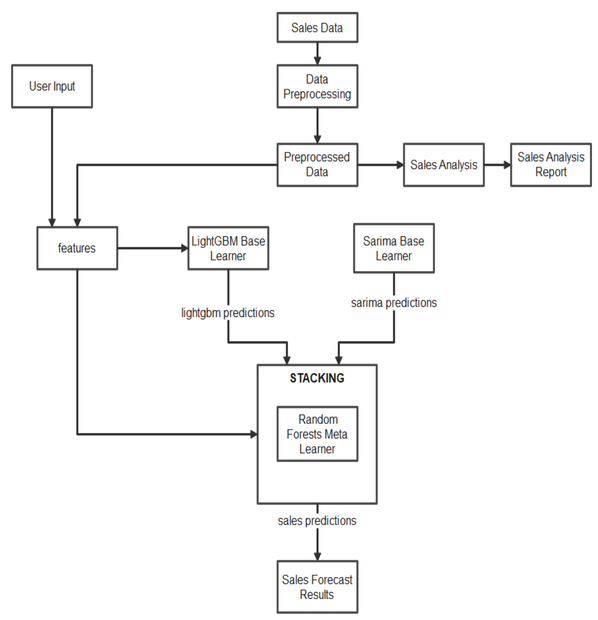
5.0. Objectives

- To develop a web-based sales forecasting system that predicts future sales by utilizing a machine learning model trained on historical sales data.
- To develop a web-based sales forecasting system that accurately predicts sales trends over a specific timeframe using historical sales data.
- To develop a web-based sales forecasting system to that analyzes sales data

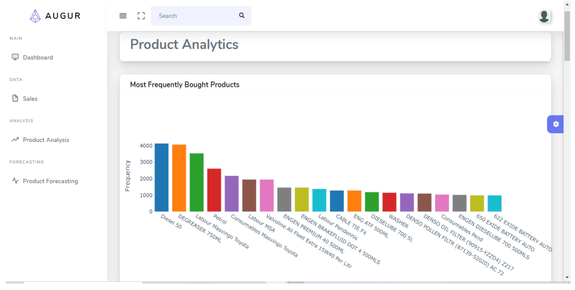
6.0. DFD DIAGRAM

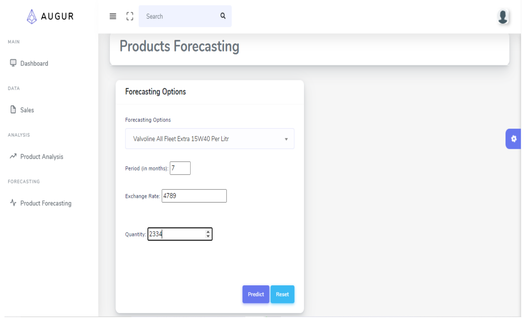


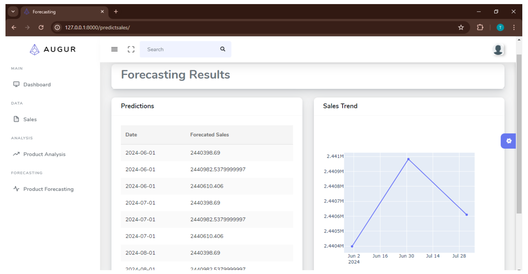
7.0. Process Flow Diagram



8.0. Screenshots







Conclusion

In conclusion, the development of a web-based sales forecasting system using a stacked ensemble model holds immense potential for the Zimbabwean retail industry. Challenges faced by local businesses in optimizing inventory management and marketing strategies in an economically volatile environment have highlighted the critical need for data-driven decision making. Traditional intuition-based planning methods have proved to lead to significant losses due to inefficiencies in addressing complexities of the market. This system addresses the critical need for data driven insights by leveraging machine learning algorithms to generate robust and accurate sales forecasts. The system utilizes historical sales data, exchange rates and seasonality to come up with accurate sales forecasts. This integration of machine learning algorithms allows for the identification of patterns, complex relationships and seasonality in sales data which enables businesses to anticipate demand, identify growth opportunities and mitigate risks of inventory management effectively. The development of a user-friendly web-application allows for the easy use of the system and the use of data visualization tools which would the easy interpretation of the results. Overall, this proposed sales forecasting system using a stacked ensemble model offers a promising solution for Zimbabwean retail businesses to thrive in an economically volatile and competitive environment.

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